

Characterization and Properties of Polyphthalamide/ Polyamide Blends and Polyphthalamide/Polyamide/ Polyolefin Blends

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Polyphthalamide(PPA)/polyamide(PA) blends were analyzed to determine the effect of PA addition to PPA on melting point, glass transition temperature, dynamic modulus, and heat distortion temperature. Results indicate that the choice of PPA and the choice of PA for the blend systems affects not only the above properties but also the miscibility of the blend systems. In general, PA addition to PPA lowers the melting point and glass transition, which potentially makes these blend materials easier to process. Also, the PPA/PA blends were observed to have dynamic modulus curves with transitions shifted to lower temperatures and crystalline plateaus shifted to lower modulus.

PPA/PA/polyolefin(PP) systems were investigated to determine if a useful balance of properties could be obtained, even though the blend components would have to be processed at unusually high processing temperatures (in excess of 320°C). Morphological characterization indicates that small dispersed domains of PP are obtained. The modifiers utilized in these systems were either found at the interface of the PP domain or dispersed within the PP domain. The properties of PPA/PA/PP blends indicates that these systems are ductile and have a good balance of strength, stiffness, impact, and thermal performance.

INTRODUCTION

In recent years, a relatively new class of crystalline polyamides has been commercialized by several polymer resin manufacturers. These materials, known as polyphthalamides [also referred to as aromatic or semi-aromatic polyamides (1, 2)] contain an aromatic group or groups in the polymer backbone. In addition to the aromatic group(s) in the polymer backbone, polyphthalamides (hereafter referred to as PPA) are typically distinguished from aliphatic polyamides (hereafter referred to as PA) by their higher melting points, higher glass transition temperatures, lower moisture absorption, and greater dimensional stability. Additionally, because of the aromaticity of the polymer backbone, PPAs typically have higher tensile strength and higher heat distortion temperatures compared to PAs.

Limited work has been reported on blends of PPA with PA (1-4). The focus of this research has been on the effect of structure of the PPA or PA on the miscibility of PPA/PA system. Physical property data for the various combinations of PPA with PA are not reported in the literature. While there have been extensive investigations into PA/polyolefin blends (e.g. 5-7), little is known about the properties of PPA/polyolefin blends (8).

Presumably this is due to the extremely high processing temperatures needed for PPA materials.

In this work, PPA/PA blends were investigated in an effort to determine how the addition of PA affects the processability (melting point) and mechanical properties (dynamic modulus, HDT) of such blends. In addition, PPA/PA/polyolefin blends are also characterized and demonstrate that a useful balance of properties can be obtained in such systems.

EXPERIMENTAL

The two polyphthalamides used in this study were supplied by Amoco Polymers, Inc. PPA "A" is Amodel® A-1000 resin - reported $T_g = 127^\circ\text{C}$, $T_m = 315^\circ\text{C}$, PPA "B" is Amodel® A-4000 resin-reported $T_g = 95^\circ\text{C}$, $T_m = 325^\circ\text{C}$. The aliphatic polyamides used in the study were DuPont Zytel® 101 (referred to as PA 6,6) and AlliedSignal Capron® 8202 (referred to as PA 6). The polyolefin was Amoco polypropylene grade 5013 (referred to as PP). The maleic anhydride functionalized modifiers were Shell Kraton® FG1901X (referred to as SEBS-g-MA), Uniroyal Royaltuf® 465 (referred to as EPDM-g-MA), and Exxon Exxelor® PO1015 (referred to as PP-g-MA). All blends were compounded on a 25 mm Berstorff co-rotating twin-screw extruder. Prior to

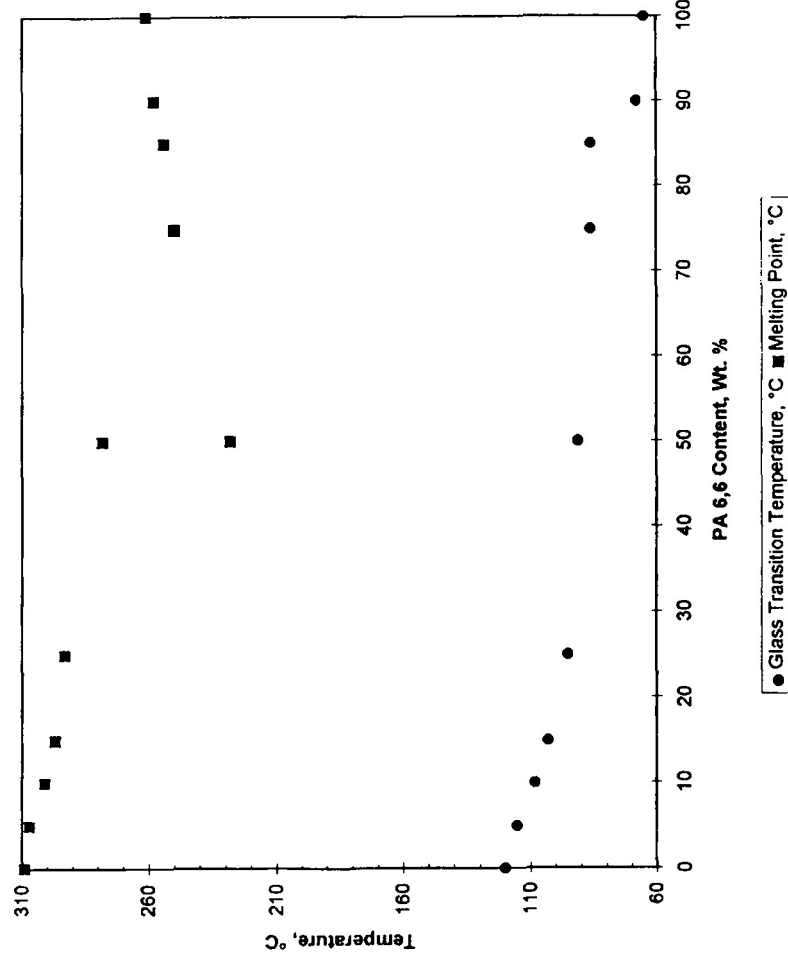


Fig. 1. Effect of PA 6,6 content on thermal transitions of PPA "A"/PA 6,6 blends.

processing the PPA, PA 6,6, and PA 6 were dried in a desiccant drier at 100, 80, & 50°C respectively for a minimum of 6 hours. Test specimens were molded on

a 25 ton Arburg All-rounder injection molding machine. Test specimens were tested dry as molded in accordance to ASTM test methods.

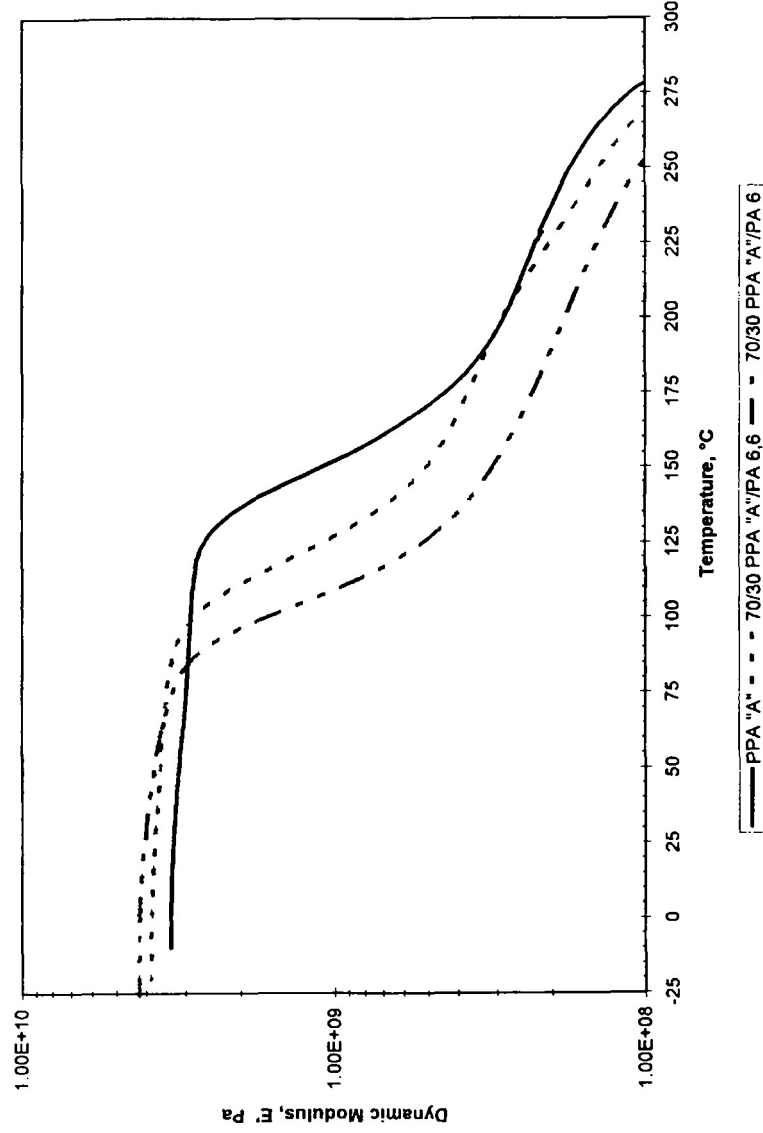


Fig. 2. Effect of PA 6,6 and PA 6 on dynamic modulus of PPA "A".

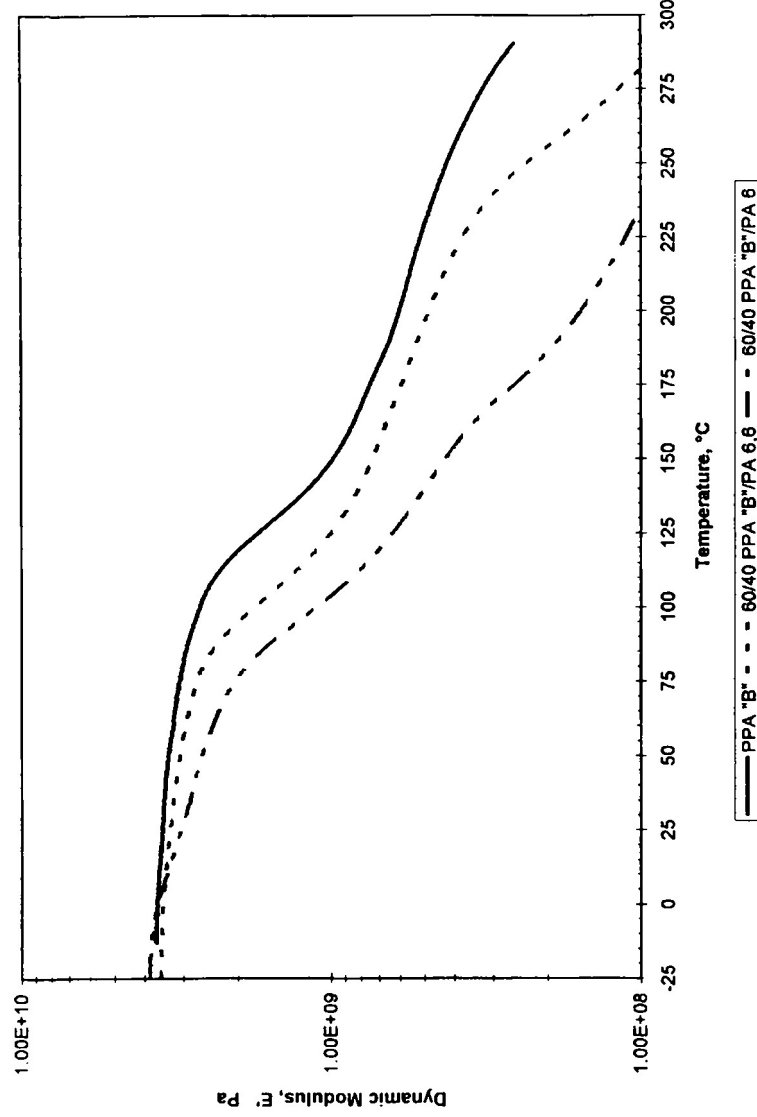


Fig. 3. Effect of PA 6.6 and PA 6 on dynamic modulus of PPA "B".

RESULTS AND DISCUSSION

Polyphthalamide/Aliphatic Polyamide (PPA/PA) Blends

Thermal analysis techniques (DSC) were utilized to assess the effects of PA 6.6 content on the melting point (T_m) and glass transition temperature (T_g) of the resultant PPA "A"/PA 6,6 blends. The data indicate that one T_g is observed over the entire composition range (Fig. 1). Generally one melting point was ob-

served over the composition range. The exception was the 50/50 composition where two distinct melting points were observed. The single melting peak for the blend compositions was broader than the melting peak for either the neat PPA "A" or PA 6.6.

The data suggest that small amounts (up to 25%) of PA 6.6 can be added to PPA "A" to make the material more processable as evidenced by the lower T_m and lower T_g . In fact, the addition of 15–25% PA 6.6 to PPA "A" lowers the T_g to a point ($<100^\circ\text{C}$) that it would be

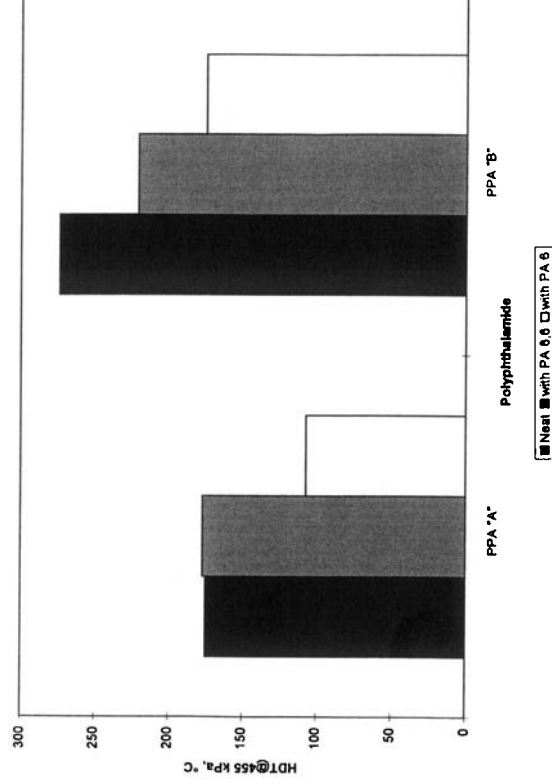


Fig. 4. HDT@455 kPa of polyphthalamide/polyamide blends.

feasible to utilize hot water molds and still achieve a product which is fully crystalline. Typically, to get fully crystalline parts, mold temperature must be at or above the T_g of the resin.

The effects of adding 30% PA 6,6 or PA 6 to PPA "A" on the crystallinity, as determined by dynamic modulus measurements, were evaluated (Fig. 2). PPA "B", which is known to have higher crystallinity than PPA "A" was also investigated for the effect of adding 40% PA 6,6 or PA 6 to PPA "B" (Fig. 3). The blends of PA 6 or PA 6,6 with PPA "A" or PPA "B" are observed to have lower T_g and lower modulus values for the crystalline plateau compared to the neat PPA "A" or PPA "B". The lower crystalline plateau for the blends indicates that there is some type of disruption in the crystalline structure or ultimate crystallinity compared to the neat materials. In the PPA "A" system only one transition is readily observed while in the PPA "B" transition several transitions are observed. This may indicate that while PA 6,6 and PA 6 are miscible with PPA "A", PA 6,6 and PA 6 may be immiscible or only partially miscible with PPA "B". In the PPA "B"/PA 6,6 and PPA "B"/PA 6 blends, the higher temperature transitions are presumed to be due to the melting out of the PA 6 or PA 6,6 crystal structure. In the PPA "A"/PA 6,6 or PA 6 blends, since no higher temperature transitions are observed, it is believed that the PA 6,6 or PA 6 have co-crystallized within the PPA "A" crystal structure. This data in conjunction with the DSC curves for these systems suggest that for PPA "B", PA 6 and PA 6,6 are immiscible while for PPA "A", PA 6 and PA 6,6 are miscible.

The HDT@455kPa was measured in an effort to relate the dynamic modulus data to a useful design property. The HDT data for the blends of PPA "A" with 30% PA 6,6 and PA 6 and the HDT data for blends of PPA "B" with 40% PA 6,6 and PA 6 display the same trends as was evidenced in the dynamic modulus data (Fig. 4). The PPA "A"/PA 6,6 blend did not exhibit a lower HDT value than the neat PPA "A" system while the PPA "A"/PA 6 blend was observed to have a lower HDT value than either the neat PPA "A" or the PPA "A"/PA 6,6 blend. (These results track the relative crystalline plateaus of each of the materials.) In the PPA "B" system, both the PPA "B"/PA 6,6 and PPA "B"/PA 6 HDT values were observed to be lower than that of the neat PPA "B".

The investigations into PPA/PA blends has revealed that PA can be added to PPA to help reduce processing temperatures (blend has lower T_m). One must be careful as to the choice of PPA and PA as these results indicate that PA 6,6 and PA 6, may not be miscible with all PPA. Additionally, this work also found that the addition of PA to PPA does tend to disrupt the crystal structure and crystallinity of the PPA, especially if the PPA is highly crystalline to start with.

Polyphthalamide/Polyamide/Polyolefin Blends

Typically, polyamides are incompatible with polyolefins and a compatibilizer must be utilized to obtain

a resulting material with useful properties. A common scheme to compatibilize these types of blends is to add an amount of polyolefin to which a functional group has been grafted. Typically, maleic anhydride is utilized and the anhydride grafted on the polyolefin backbone can react with the amine end groups of the polyamide to form a block copolymer in situ. This block copolymer can then act as the compatibilization agent. One of the difficulties associated with polyphthalamide/polyolefin blends is the high temperatures that are needed to process the polyphthalamides and which the polyolefin, functional polyolefin, and in situ block copolymer must be subjected to. (Typically, PPA is processed at temperatures in excess of 320°C.) By utilizing a combination of process stabilizers, extruder screw design, and extruder processing sequences and operating conditions, compounded PPA/PA/polyolefin blends were successfully manufactured.

Morphological characterization of the PPA/PA/polyolefin systems was similar to morphology reported in PA/polyolefin systems (7). PP domain sizes were found to be on the order of 1 μ m. A unique feature that was found in the PPA/PA/PP/PP-g-MA/SEBS-g-MA system was a core-shell morphology in which the SEBS-g-MA was clearly found either as discrete particles or only at the interface between the PP/PP-g-MA domain and the PPA/PA phase (Fig. 5). This type of morphology was not seen when EPDM-g-MA was utilized as the modifier. In that system, EPDM-g-MA was found as discrete particles or as particles within the

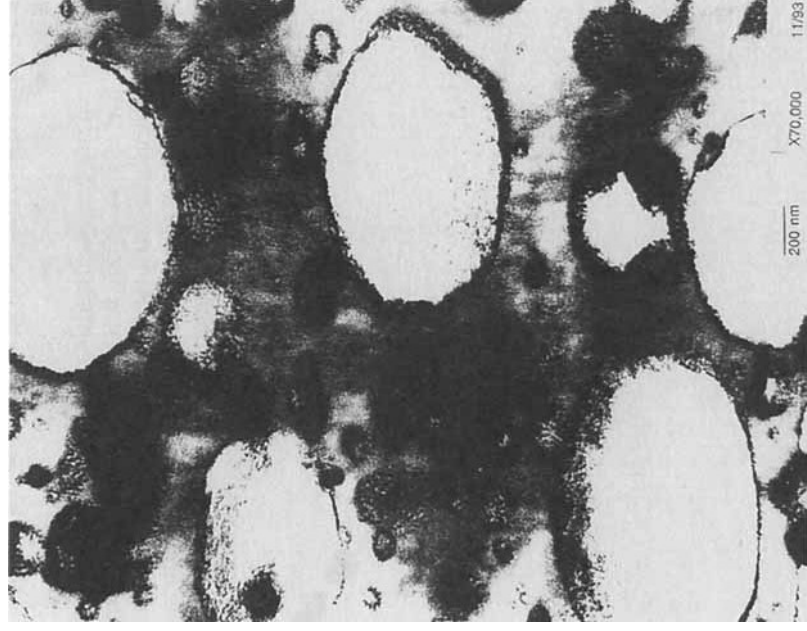


Fig. 5. Morphology of 75/12.5/5/7.5 PPA "B"/PP/PP-g-MA/SEBS-g-MA blend.

Table 1. Effect of PA 6,6 on Melting Point of PPA "A"/Polyolefin Blends.

PPA "A" Content	55	61.25	55	50	50	45	70	40	30
PA 6,6 Content	0	8.75	10	10	15	15	30	20	45
Total PA Content	55	70	65	60	65	60	100	60	75
Polyolefin/Modifier Content	45	30	35	40	35	40	0	40	25
Wt. % PA 6,6 in PA phase	0	12.5	15.38	16.67	23.08	25	30	33.33	60
Melting Point, °C	319	317	317	318	313	312	310	308	322

PP/PP-g-MA phase. No evidence of EPDM-g-MA at the interface between PP/PP-g-MA phase and the PPA/PA phase was found.

As was seen in the PPA/PA blends, the addition of PA 6,6 lowers the melting point of these multicomponent blend systems when added at a loading of less than 50% (Table 1). At a loading of 60% PA 6,6 two distinct melting points are observed (similar to the 50/50 PPA "A"/PA 6,6 results reported above). It would appear that the addition of PA 6,6 to PPA "A"/polyolefin blends is a viable means of generating blends with lower processing temperatures.

Selected formulations were manufactured with either PPA "A" or PPA "B" and PA 6,6 or PA 6 along with PP, PP-g-MA, SEBS-g-MA and/or EPDM-g-MA (Table 2). The combination of tensile properties, flexural properties, impact properties, and HDT can be used to judge the usefulness of such compositions. All blends were found to be ductile as evidenced by the elongation as break. As was previously noted, PPA "A" and PPA "B" are generally brittle in the neat state with an elongation at break of generally 4–5%.

The blend systems in which SEBS-g-MA was utilized instead of EPDM-g-MA, generally exhibited a better balance of properties as evidenced by higher impact strengths while maintaining essentially the same strength and stiffness of comparable EPDM-g-MA systems. (Compare 3 vs. 2, 5 vs. 4). One system, (5 in Table 2), exhibited supertoughness even though the loading of SEBS-g-MA modifier (12.5%) is generally

lower than what is typically needed for supertoughness in PPA (generally 15–20% SEBS-g-MA is needed).

The physical property data also indicates that producing blends made with PPA "B" was successful, even though PPA "B" is typically processed at temperatures in excess of 340°C. The property data indicate that PPA "B"/PA/polyolefin blends exhibit good ductility, good flexural properties, and higher HDT@455 kPa than blends made from PPA "A". The higher HDT is due to the higher crystallinity of PPA "B" compared to PPA "A" while the ductility and stiffness differences are related to the choice of PA and modifier and the loadings of each.

As can be seen from the properties of the selected systems, PPA/PA/polyolefin blends are ductile and exhibit a good balance of strength, stiffness, and toughness. The combination of ductility and modulus for these unfilled blends indicates that these types of materials will have good "practical" toughness. Since the continuous phase in each of these blends is the PPA/PA combination, expectations are that these types of materials will also have good dimensional stability, low moisture adsorption and good elevated temperature performance. One can see from the HDT values at both 455 kPa and 1820 kPa that these types of blend systems can be used at temperatures higher than is typically used for polyamides, polyolefins, or polyamide/polyolefin blends. Physical properties from these types of blend systems will ultimately be determined by the choice of PPA, the choice of PA, and the

Table 2. Properties of Selected Polyphthalamide/Polyamide/Polyolefin Blends.

	1	2	3	4	5	6	7	8	9	10
PPA "A"	40	40	40	65.625	65.625	—	—	—	—	—
PPA "B"	—	—	—	—	—	36	65	36	36	65.625
PA 6,6	—	20	20	9.375	9.375	—	—	24	24	9.375
PA 6	20	—	—	—	—	24	10	—	—	—
PP	30	30	30	7.5	7.5	35	15	30	27.5	7.5
PP-g-MA	5	5	5	5	5	5	5	5	5	5
SEBS-g-MA	—	—	5	—	12.5	—	10	—	—	—
EPDM-g-MA	5	5	—	12.5	—	—	—	5	7.5	12.5
Tensile strength, MPa	56.9	47.5	52.7	64.0	65.4	58.8	59.2	55.5	50.8	59.7
Tensile elongation at break, %	16.7	9.5	12.2	40.0	42.1	13.2	23.1	21.0	35.8	27.2
Flexural strength, MPa	86.8	71.9	77.2	89.7	90.1	91.0	89.1	81.8	69.8	85.7
Flexural modulus, MPa	2199	2025	1855	1965	1997	2427	2116	2146	1905	2028
Notched Izod impact, J/m	69	69	102	194	1164	48	117	48	105	102
HDT @455 kPa, °C	135	126	113	—	—	158	158	176	184	—
HDT @1820 kPa, °C	—	91	87	—	—	—	—	—	83	—

choice of modifier. Having these choices allows one to selectively design and tailor the blend for the properties desired.

CONCLUSIONS

This work has shown that the addition of PA to PPA can form either miscible or immiscible blend systems, depending on the choice of PPA and PA. Additionally, the PPA/PA blends exhibit lower melting points and glass transition temperatures compared to the neat PPA, suggesting that the incorporation of PA into PPA is a means to improve the processability of PPA resins. PPA/PA/polyolefin blends were successfully produced even though processing temperatures for PPA's are generally in excess of 320°C. Two different blend morphologies were observed for these types of systems. For blends made with SEBS-g-MA, a core-shell morphology was observed with SEBS-g-MA found either alone or at the interface of the PP/PP-g-MA domains. In the EPDM-g-MA systems, EPDM-g-MA was found either alone or dispersed within the PP/PP-g-MA

phase. In either case, the PPA/PA/polyolefin blends exhibited a good balance of ductility, strength, stiffness, impact, and thermal performance, indicating that these types of materials should have commercial utility.

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REFERENCES

1. T. S. Ellis, *Polymer*, **29**, 2015 (1988).
2. T. S. Ellis, *Polymer*, **31**, 1058 (1990).
3. T. S. Ellis, *Macromolecules*, **24**, 3845 (1991).
4. T. S. Ellis, *Polym. Commun.*, **32**, no. 32, p. 489 (1991).
5. R. G. Lutz *et al.*, U.S. Patent 4,795,782 (1989).
6. R. G. Lutz *et al.*, U.S. Patent 5,006,601 (1991).
7. M. J. Modic and L. A. Pottick, *SPE ANTEC Tech. Papers*, **37**, 1907 (1991).
8. G. P. Desio *et al.*, U.S. Patent 5,436,294 (1995).